

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]		opcode		B-type
imm[31:12]										rd		opcode		U-type
imm[20 10:1 11 19:12]										rd		opcode		J-type

RV32I Base Instruction Set

imm[31:12]					rd	0110111	LUI	
imm[31:12]					rd	0010111	AUIPC	
imm[20 10:1 11 19:12]					rd	1101111	JAL	
imm[11:0]			rs1	000	rd	1100111	JALR	
imm[12 10:5]		rs2	rs1	000	imm[4:1 11]	1100011	BEQ	
imm[12 10:5]		rs2	rs1	001	imm[4:1 11]	1100011	BNE	
imm[12 10:5]		rs2	rs1	100	imm[4:1 11]	1100011	BLT	
imm[12 10:5]		rs2	rs1	101	imm[4:1 11]	1100011	BGE	
imm[12 10:5]		rs2	rs1	110	imm[4:1 11]	1100011	BLTU	
imm[12 10:5]		rs2	rs1	111	imm[4:1 11]	1100011	BGEU	
imm[11:0]			rs1	000	rd	0000011	LB	
imm[11:0]			rs1	001	rd	0000011	LH	
imm[11:0]			rs1	010	rd	0000011	LW	
imm[11:0]			rs1	100	rd	0000011	LBU	
imm[11:0]			rs1	101	rd	0000011	LHU	
imm[11:5]		rs2	rs1	000	imm[4:0]	0100011	SB	
imm[11:5]		rs2	rs1	001	imm[4:0]	0100011	SH	
imm[11:5]		rs2	rs1	010	imm[4:0]	0100011	SW	
imm[11:0]			rs1	000	rd	0010011	ADDI	
imm[11:0]			rs1	010	rd	0010011	SLTI	
imm[11:0]			rs1	011	rd	0010011	SLTIU	
imm[11:0]			rs1	100	rd	0010011	XORI	
imm[11:0]			rs1	110	rd	0010011	ORI	
imm[11:0]			rs1	111	rd	0010011	ANDI	
0000000		rs2	rs1	000	rd	0110011	ADD	
0100000		rs2	rs1	000	rd	0110011	SUB	
0000000		rs2	rs1	001	rd	0110011	SLL	
0000000		rs2	rs1	010	rd	0110011	SLT	
0000000		rs2	rs1	011	rd	0110011	SLTU	
0000000		rs2	rs1	100	rd	0110011	XOR	
0000000		rs2	rs1	101	rd	0110011	SRL	
0100000		rs2	rs1	101	rd	0110011	SRA	
0000000		rs2	rs1	110	rd	0110011	OR	
0000000		rs2	rs1	111	rd	0110011	AND	
fm	pred		succ	rs1	000	rd	0001111	FENCE
0000000		00000	00000	000	00000	1110011	ECALL	
0000000		00001	00000	000	00000	1110011	EBREAK	
0000000		shamt	rs1	001	rd	0010011	SLLI	
0000000		shamt	rs1	101	rd	0010011	SRLI	
0100000		shamt	rs1	101	rd	0010011	SRAI	
1000001		10011	rs1	1000	rd	0001111	FENCE.TSO	

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV64I Base Instruction Set (in addition to RV32I)

imm[11:0]						rs1		110		rd		0000011			LWU
imm[11:0]						rs1		011		rd		0000011			LD
imm[11:5]				rs2		rs1		011		imm[4:0]		0100011			SD
000000				shamt		rs1		001		rd		0010011			SLLI
000000				shamt		rs1		101		rd		0010011			SRLI
010000				shamt		rs1		101		rd		0010011			SRAI
imm[11:0]						rs1		000		rd		0011011			ADDIW
0000000				shamt		rs1		001		rd		0011011			SLLIW
0000000				shamt		rs1		101		rd		0011011			SRLIW
0100000				shamt		rs1		101		rd		0011011			SRAIW
0000000				rs2		rs1		000		rd		0111011			ADDW
0100000				rs2		rs1		000		rd		0111011			SUBW
0000000				rs2		rs1		001		rd		0111011			SLW
0000000				rs2		rs1		101		rd		0111011			SRLW
0100000				rs2		rs1		101		rd		0111011			SRAW

RV32/RV64 Zifencei Standard Extension

imm[11:0]						rs1		001		rd		0001111			FENCE.I
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RV32/RV64 Zicsr Standard Extension

csr						rs1		001		rd		1110011			CSRRW
csr						rs1		010		rd		1110011			CSRRS
csr						rs1		011		rd		1110011			CSRRC
csr						zimm5		101		rd		1110011			CSRRWI
csr						zimm5		110		rd		1110011			CSRRSI
csr						zimm5		111		rd		1110011			CSRRCI

RV32M Standard Extension

0000001				rs2		rs1		000		rd		0110011			MUL
0000001				rs2		rs1		001		rd		0110011			MULH
0000001				rs2		rs1		010		rd		0110011			MULHSU
0000001				rs2		rs1		011		rd		0110011			MULHU
0000001				rs2		rs1		100		rd		0110011			DIV
0000001				rs2		rs1		101		rd		0110011			DIVU
0000001				rs2		rs1		110		rd		0110011			REM
0000001				rs2		rs1		111		rd		0110011			REMU

RV64M Standard Extension (in addition to RV32M)

0000001				rs2		rs1		000		rd		0111011			MULW
0000001				rs2		rs1		100		rd		0111011			DIVW
0000001				rs2		rs1		101		rd		0111011			DIVUW
0000001				rs2		rs1		110		rd		0111011			REMW
0000001				rs2		rs1		111		rd		0111011			REMUW

31	27	26	25	24	20	19	15	14	12	11	7	6	0
funct7				rs2		rs1		funct3		rd		opcode	

R-type

RV32A Standard Extension

00010	aq	rl	00000	rs1	010	rd	0101111	LR.W
00011	aq	rl	rs2	rs1	010	rd	0101111	SC.W
00001	aq	rl	rs2	rs1	010	rd	0101111	AMOSWAP.W
00000	aq	rl	rs2	rs1	010	rd	0101111	AMOADD.W
00100	aq	rl	rs2	rs1	010	rd	0101111	AMOXOR.W
01100	aq	rl	rs2	rs1	010	rd	0101111	AMOAND.W
01000	aq	rl	rs2	rs1	010	rd	0101111	AMOOR.W
10000	aq	rl	rs2	rs1	010	rd	0101111	AMOMIN.W
10100	aq	rl	rs2	rs1	010	rd	0101111	AMOMAX.W
11000	aq	rl	rs2	rs1	010	rd	0101111	AMOMINU.W
11100	aq	rl	rs2	rs1	010	rd	0101111	AMOMAXU.W

RV64A Standard Extension (in addition to RV32A)

00010	aq	rl	00000	rs1	011	rd	0101111	LR.D
00011	aq	rl	rs2	rs1	011	rd	0101111	SC.D
00001	aq	rl	rs2	rs1	011	rd	0101111	AMOSWAP.D
00000	aq	rl	rs2	rs1	011	rd	0101111	AMOADD.D
00100	aq	rl	rs2	rs1	011	rd	0101111	AMOXOR.D
01100	aq	rl	rs2	rs1	011	rd	0101111	AMOAND.D
01000	aq	rl	rs2	rs1	011	rd	0101111	AMOOR.D
10000	aq	rl	rs2	rs1	011	rd	0101111	AMOMIN.D
10100	aq	rl	rs2	rs1	011	rd	0101111	AMOMAX.D
11000	aq	rl	rs2	rs1	011	rd	0101111	AMOMINU.D
11100	aq	rl	rs2	rs1	011	rd	0101111	AMOMAXU.D

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32F Standard Extension

imm[11:0]			rs1	010	rd	0000111	FLW
imm[11:5]		rs2	rs1	010	imm[4:0]	0100111	FSW
rs3	00	rs2	rs1	rm	rd	1000011	FMADD.S
rs3	00	rs2	rs1	rm	rd	1000111	FMSUB.S
rs3	00	rs2	rs1	rm	rd	1001011	FNMSUB.S
rs3	00	rs2	rs1	rm	rd	1001111	FNMADD.S
0000000		rs2	rs1	rm	rd	1010011	FADD.S
0000100		rs2	rs1	rm	rd	1010011	FSUB.S
0001000		rs2	rs1	rm	rd	1010011	FMUL.S
0001100		rs2	rs1	rm	rd	1010011	FDIV.S
0101100		00000	rs1	rm	rd	1010011	FSQRT.S
0010000		rs2	rs1	000	rd	1010011	FSGNJ.S
0010000		rs2	rs1	001	rd	1010011	FSGNJN.S
0010000		rs2	rs1	010	rd	1010011	FSGNJX.S
0010100		rs2	rs1	000	rd	1010011	FMIN.S
0010100		rs2	rs1	001	rd	1010011	FMAX.S
1100000		00000	rs1	rm	rd	1010011	FCVT.W.S
1100000		00001	rs1	rm	rd	1010011	FCVT.WU.S
1110000		00000	rs1	000	rd	1010011	FMV.X.W
1010000		rs2	rs1	010	rd	1010011	FEQ.S
1010000		rs2	rs1	001	rd	1010011	FLT.S
1010000		rs2	rs1	000	rd	1010011	FLE.S
1110000		00000	rs1	001	rd	1010011	FCLASS.S
1101000		00000	rs1	rm	rd	1010011	FCVT.S.W
1101000		00001	rs1	rm	rd	1010011	FCVT.S.WU
1111000		00000	rs1	000	rd	1010011	FMV.W.X
0000000		00001	00000	010	rd	1110011	FRFLAGS
0000000		00001	rs1	001	rd	1110011	FSFLAGS
0000000		00001	zimm5	101	rd	1110011	FSFLAGSI
0000000		00010	00000	010	rd	1110011	FRRM
0000000		00010	rs1	001	rd	1110011	FSRM
0000000		00010	zimm5	101	rd	1110011	FSRMI
0000000		00011	rs1	001	rd	1110011	FSCSR
0000000		00011	00000	010	rd	1110011	FRCSR

RV64F Standard Extension (in addition to RV32F)

1100000				00010	rs1	rm	rd	1010011	FCVT.L.S
1100000				00011	rs1	rm	rd	1010011	FCVT.LU.S
1101000				00010	rs1	rm	rd	1010011	FCVT.S.L
1101000				00011	rs1	rm	rd	1010011	FCVT.S.LU

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32D Standard Extension

imm[11:0]			rs1	011	rd	0000111	FLD
imm[11:5]		rs2	rs1	011	imm[4:0]	0100111	FSD
rs3	01	rs2	rs1	rm	rd	1000011	FMADD.D
rs3	01	rs2	rs1	rm	rd	1000111	FMSUB.D
rs3	01	rs2	rs1	rm	rd	1001011	FNMSUB.D
rs3	01	rs2	rs1	rm	rd	1001111	FNMADD.D
0000001		rs2	rs1	rm	rd	1010011	FADD.D
0000101		rs2	rs1	rm	rd	1010011	FSUB.D
0001001		rs2	rs1	rm	rd	1010011	FMUL.D
0001101		rs2	rs1	rm	rd	1010011	FDIV.D
0101101		00000	rs1	rm	rd	1010011	FSQRT.D
0010001		rs2	rs1	000	rd	1010011	FSGNJ.D
0010001		rs2	rs1	001	rd	1010011	FSGNJN.D
0010001		rs2	rs1	010	rd	1010011	FSGNJX.D
0010101		rs2	rs1	000	rd	1010011	FMIN.D
0010101		rs2	rs1	001	rd	1010011	FMAX.D
0100000		00001	rs1	rm	rd	1010011	FCVT.S.D
0100001		00000	rs1	rm	rd	1010011	FCVT.D.S
1010001		rs2	rs1	010	rd	1010011	FEQ.D
1010001		rs2	rs1	001	rd	1010011	FLT.D
1010001		rs2	rs1	000	rd	1010011	FLE.D
1110001		00000	rs1	001	rd	1010011	FCLASS.D
1100001		00000	rs1	rm	rd	1010011	FCVT.W.D
1100001		00001	rs1	rm	rd	1010011	FCVT.WU.D
1101001		00000	rs1	rm	rd	1010011	FCVT.D.W
1101001		00001	rs1	rm	rd	1010011	FCVT.D.WU

RV64D Standard Extension (in addition to RV32D)

1100001		00010		rs1		rm		rd		1010011			FCVT.L.D
1100001		00011		rs1		rm		rd		1010011			FCVT.LU.D
1110001		00000		rs1		000		rd		1010011			FMV.X.D
1101001		00010		rs1		rm		rd		1010011			FCVT.D.L
1101001		00011		rs1		rm		rd		1010011			FCVT.D.LU
1111001		00000		rs1		000		rd		1010011			FMV.D.X

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]				rs2		rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32Q Standard Extension

imm[11:0]			rs1	100	rd	0000111	FLQ
imm[11:5]		rs2	rs1	100	imm[4:0]	0100111	FSQ
rs3	11	rs2	rs1	rm	rd	1000011	FMADD.Q
rs3	11	rs2	rs1	rm	rd	1000111	FMSUB.Q
rs3	11	rs2	rs1	rm	rd	1001011	FNMSUB.Q
rs3	11	rs2	rs1	rm	rd	1001111	FNMADD.Q
0000011		rs2	rs1	rm	rd	1010011	FADD.Q
0000111		rs2	rs1	rm	rd	1010011	FSUB.Q
0001011		rs2	rs1	rm	rd	1010011	FMUL.Q
0001111		rs2	rs1	rm	rd	1010011	FDIV.Q
0101111		00000	rs1	rm	rd	1010011	FSQRT.Q
0010011		rs2	rs1	000	rd	1010011	FSGNJ.Q
0010011		rs2	rs1	001	rd	1010011	FSGNJN.Q
0010011		rs2	rs1	010	rd	1010011	FSGNJX.Q
0010111		rs2	rs1	000	rd	1010011	FMIN.Q
0010111		rs2	rs1	001	rd	1010011	FMAX.Q
0100000		00011	rs1	rm	rd	1010011	FCVT.S.Q
0100011		00000	rs1	rm	rd	1010011	FCVT.Q.S
0100001		00011	rs1	rm	rd	1010011	FCVT.D.Q
0100011		00001	rs1	rm	rd	1010011	FCVT.Q.D
1010011		rs2	rs1	010	rd	1010011	FEQ.Q
1010011		rs2	rs1	001	rd	1010011	FLT.Q
1010011		rs2	rs1	000	rd	1010011	FLE.Q
1110011		00000	rs1	001	rd	1010011	FCLASS.Q
1100011		00000	rs1	rm	rd	1010011	FCVT.W.Q
1100011		00001	rs1	rm	rd	1010011	FCVT.WU.Q
1101011		00000	rs1	rm	rd	1010011	FCVT.Q.W
1101011		00001	rs1	rm	rd	1010011	FCVT.Q.WU

RV64Q Standard Extension (in addition to RV32Q)

1100011				00010	rs1	rm	rd	1010011	FCVT.L.Q
1100011				00011	rs1	rm	rd	1010011	FCVT.LU.Q
1101011				00010	rs1	rm	rd	1010011	FCVT.Q.L
1101011				00011	rs1	rm	rd	1010011	FCVT.Q.LU

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]				rs1		funct3		rd		opcode		I-type		
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32Zfh Standard Extension

rs3	10	rs2	rs1	rm	rd	1000011	FMADD.H
rs3	10	rs2	rs1	rm	rd	1000111	FMSUB.H
rs3	10	rs2	rs1	rm	rd	1001011	FNMSUB.H
rs3	10	rs2	rs1	rm	rd	1001111	FNMADD.H
0000010		rs2	rs1	rm	rd	1010011	FADD.H
0000110		rs2	rs1	rm	rd	1010011	FSUB.H
0001010		rs2	rs1	rm	rd	1010011	FMUL.H
0001110		rs2	rs1	rm	rd	1010011	FDIV.H
0101110		00000	rs1	rm	rd	1010011	FSQRT.H
0010010		rs2	rs1	000	rd	1010011	FSGNJ.H
0010010		rs2	rs1	001	rd	1010011	FSGNJN.H
0010010		rs2	rs1	010	rd	1010011	FSGNJX.H
0010110		rs2	rs1	000	rd	1010011	FMIN.H
0010110		rs2	rs1	001	rd	1010011	FMAX.H
1010010		rs2	rs1	010	rd	1010011	FEQ.H
1010010		rs2	rs1	001	rd	1010011	FLT.H
1010010		rs2	rs1	000	rd	1010011	FLE.H
1110010		00000	rs1	001	rd	1010011	FCLASS.H
1100010		00000	rs1	rm	rd	1010011	FCVT.W.H
1100010		00001	rs1	rm	rd	1010011	FCVT.WU.H
1101010		00000	rs1	rm	rd	1010011	FCVT.H.W
1101010		00001	rs1	rm	rd	1010011	FCVT.H.WU

RV64Zfh Standard Extension (in addition to RV32Zfh)

1100010		00010	rs1	rm	rd	1010011	FCVT.L.H
1100010		00011	rs1	rm	rd	1010011	FCVT.LU.H
1101010		00010	rs1	rm	rd	1010011	FCVT.H.L
1101010		00011	rs1	rm	rd	1010011	FCVT.H.LU

Table 1: Instruction listing for RISC-V